Assignment 5

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Econometrics I
NYU Stern

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Instructions. Two problems on panel event studies and staggered difference-in-differences (Sessions 8–9). `R` with `fixest` (which has `i()`, `iplot()`, and `sunab()`) or Python with `pyfixest`. Turn in a PDF with your results and reasoning, and your code. Derive first, then simulate to check yourself.

1 Staggered adoption of a pricing tool

600 firms observed for 16 quarters. Firms adopt an algorithmic-pricing tool in one of three cohorts, quarter 5, 9, or 13, or never; assign each firm to one of the four groups at random with equal probability. Log gross margin is

$$y_{it} = \alpha_i + \gamma_t + \tau_{it} + \varepsilon_{it}, \qquad \alpha_i \sim \mathcal{N}(0, 0.3^2),\ \varepsilon_{it} \sim \mathcal{N}(0, 0.05^2),$$

with γ_t a random walk with drift ($\gamma_t = \gamma_{t-1} + 0.01 + 0.02\,\zeta_t$) common to all firms. The treatment effect ramps in, at a speed and to a level that differ by cohort:

$$\tau_{it} = \bar{\tau}_g \cdot \min\!\left(1,\ \frac{k_{it}+1}{s_g}\right) \text{ for } k_{it} = t - g \geq 0, \quad \text{and 0 before adoption,}$$

where cohort $g = 5$ has $\bar{\tau} = 0.10$, $s = 1$ (full effect immediately); $g = 9$ has $\bar{\tau} = 0.06$, $s = 2$; $g = 13$ has $\bar{\tau} = 0.03$, $s = 4$. Early adopters gain more and faster. Never-treated firms have $\tau = 0$.

(a) Simulate. Compute the true average treatment effect on the treated: the mean of τ_{it} over treated firm-quarters (post-adoption). Also report it by cohort.

(b) Estimate the static two-way fixed effects regression, y on a post-adoption dummy with firm and quarter fixed effects, clustered by firm. Compare to (a). Which cohorts is it under- or over-weighting, and why? (Think about how many post-periods each cohort contributes.)

(c) Estimate the naive event study: y on event-time dummies (reference $k = -1$; never-treated firms get the reference value) with firm and quarter fixed effects. Plot it. Run a joint test on the pre-period coefficients. They should reject even though no firm has a pre-trend. Explain, in the language of Session 9, what contaminates the pre-period coefficients here.

(d) Re-estimate with `sunab(cohort, quarter)` (Sun and Abraham 2021), plot, and compare the post-period coefficients to the true average of τ_{it} at each event time. Report the aggregated ATT and compare to (a).

(e) Drop the never-treated firms and re-run (d). What does `sunab` use as the control group now, and which event-time coefficients become unavailable? One paragraph.

(f) Managerial summary, three sentences: the analytics team reports the static two-way FE number from (b) as "the effect of the tool." What is wrong with it, what number would you report instead, and to whom does it apply?

2 Goodman-Bacon by hand

Two cohorts of equal size and three periods. The *early* cohort is treated in periods 2 and 3; the *late* cohort is treated in period 3 only. There is no never-treated group. Potential outcomes: $y_{it}(0) = \alpha_i + \gamma_t$, and the treatment effect for a unit treated for $k+1$ periods is $\tau(k)$, so the early cohort has effect $\tau(0)$ in period 2 and $\tau(1)$ in period 3, and the late cohort has effect $\tau(0)$ in period 3.

(a) Write the population mean of y for each cohort-period cell (six cells) in terms of $\bar{\alpha}_g$, γ_t, and $\tau(\cdot)$.

(b) There are two 2×2 difference-in-differences comparisons available. Comparison A uses periods 1–2 with the late cohort (not yet treated) as the control for the early cohort. Comparison B uses periods 2–3 with the *early* cohort (already treated) as the control for the late cohort. Compute what each estimates in terms of $\tau(0)$ and $\tau(1)$. One of them is not a treatment effect. Which, and what is it instead?

(c) Show that the two-way fixed effects regression of y on the treatment dummy with unit and period effects estimates $\frac{1}{2}A + \frac{1}{2}B$ in this design. (Hint: with equal cohort sizes and this timing, the within-unit-and-period residualized treatment dummy takes only a few values; the TWFE coefficient is a variance-weighted average of the 2×2s, and here the two comparisons have equal variance. You may argue from symmetry rather than grinding the algebra.)

(d) Let $\tau(0) = 0.5$ and $\tau(1) = 1.0$: the effect grows the longer a firm has the tool. What does TWFE estimate? Compare it to every true effect in the design. Now let $\tau(1) = 1.5$.

(e) Simulate the design (100 firms per cohort, small noise) and confirm (c) and (d): compute A and B from the six cell means and compare their average to `feols(y ~ D | id + t)`.

(f) One paragraph: what feature of the design makes comparison B appear, why is it harmless when $\tau(k)$ is constant in k, and what do Callaway–Sant'Anna and Sun–Abraham do about it?